

Tim Sananikone

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Education

University of California, Irvine – BS in Computer Science

GPA: 3.71

Relevant Coursework: Project in Computer Vision, Neural Networks and Deep Learning, Machine Learning and Data-Mining, Artificial Intelligence, Data Structures and Algorithms, Parallel and Distributed Computing, Embedded Software, Internet of Things (IoT) Software and Systems

Skills

Programming Languages: Python, C++, C, Java, JavaScript, Assembly, MATLAB, HTML/CSS

Technologies & Frameworks: PyTorch, OpenCV, Segment Anything Model (SAM), YOLOv8, Neural Radiance Fields (NeRF), Git/GitHub, Unix/Linux Command Line, ESP32 Programming, Parallel Programming

Technical Skills: Machine Learning, Computer Vision, Distributed Systems, Large-Scale Data Processing, Software Development, System Design, Embedded Systems

Experience

Machine Learning Intern – Monterey Bay Aquarium Research Institute (MBARI), Moss Landing, CA

- Designed large-scale deep learning pipeline with SAM and SAHI to process 26,000+ aerial images (42MP) across six coastal sites, overcoming glint, variable lighting, and irregular structures
- Developed and published CLI Python package (PyPI) for Unix/Linux environments
- Implemented GPU parallelization for concurrent inference across image slices, reducing processing time by 90%
- Evaluated multiple ML architectures including YOLOv8 and SAM for segmentation pipeline
- Managed development using Git/GitHub version control

Undergraduate Researcher – Molloy Lab, Irvine, CA

- Developed 3D U-Net for medical image segmentation with custom preprocessing and patch-based pipeline
- Built data augmentation system (flips, deformations, transformations) achieving 94% accuracy (dice score) with only 25 training volumes
- Developed digital phantom generation pipeline producing 50+ synthetic CT volumes to test multi-material decomposition algorithms, leading to an increase of iodine map clarity by 20% and 2025 AAPM abstract acceptance
- Created RAW-to-DICOM conversion tool standardizing lab workflows and improving data pipeline compatibility

Projects

Light-Sensing Plant Robot – GitHub

- Developed autonomous navigation system using ESP32 microcontroller programmed in C++ utilizing real-time sensor processing
- Implemented Hardware-in-the-Loop (HITL) testing methodology, creating simulation environment to validate navigation algorithms with replayed sensor data before deployment
- Designed software architecture integrating real-time sensor processing with motor control systems
- Managed project development using Git version control for iterative design and testing

NeRF Model for 3D Reconstruction – GitHub

- Implemented Neural Radiance Fields (NeRF) model from scratch in Python using hierarchical sampling and dual-network architecture
- Designed and optimized algorithms for volumetric rendering, achieving mean squared error of 0.0018 for high-fidelity 3D reconstructions from sparse 2D image inputs
- Optimized rendering pipeline for improved performance and visual quality through algorithmic improvements